BOLT

BOOK ORIENTED LIMIT TRADING

Pool - Peshwas

WHAT LURKS BENEATH — THE RIDDLES

01 **NUMBERS HUM WHERE NO EYE CAN SEE, THREADS TWIST IN WAYS THE MIND CANNOT TRACE. SOME PATHS CARRY WEIGHT THAT WAS NEVER GIVEN, AND ONLY THE SILENT ARCHITECT KNOWS WHICH FLOW MULTIPLIES UNSEEN.**

- Correlation

02 **I KEEP TWO LISTS: ONE OF BREATHS, ONE OF SPARKS. READ NEITHER AS THEY READ THEMSELVES, INDEX BY MY HIDDEN HAND. WHEN YOU FOLD THEIR TALLIES THROUGH THE KEY I NEVER WRITE, A SINGLE NOTION BLINKS BACK; CALL IT WHAT YOU WILL..**

- Arbitrage

03 **NOT THE RISE, NOR FALL, I CLAIM, BUT WHERE THE RISE BEGINS TO WANE. I NEVER SPEAK IN STRAIGHT OR STEEP, YET I'M THE WHISPER IN THE DEEP.**

- Acceleration

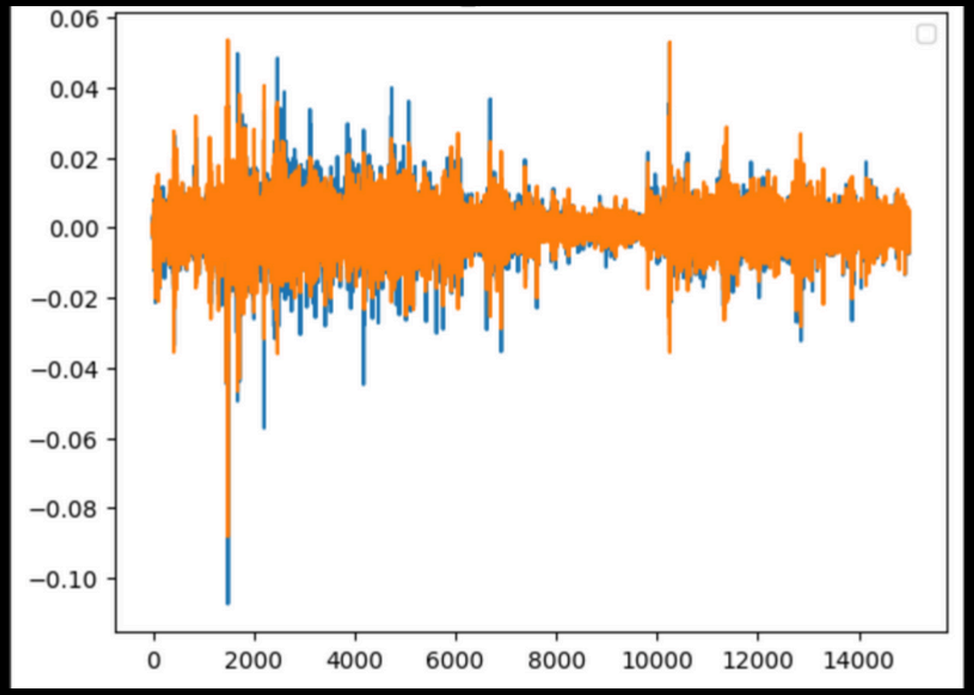
EXPLORATORY DATA ANALYSIS

01 NO TRADE ASSET CORRELATION

- NoTrade had a high correlation with asset1.
- We saw that the log of any of the order book prices of asset1 when differentiated twice and shifted upwards had a very high negative correlation with the difference of NoTrade
- Hence, we figured out a way to use this data to for our trading strategy.

```
log1=np.log(asset1['ask_px1'])
diff=log1.diff().diff().shift(-1)
correlation = diff.corr(no_trade['signal'].diff())
correlation
```

```
np.float64(-0.9454648506402126)
```

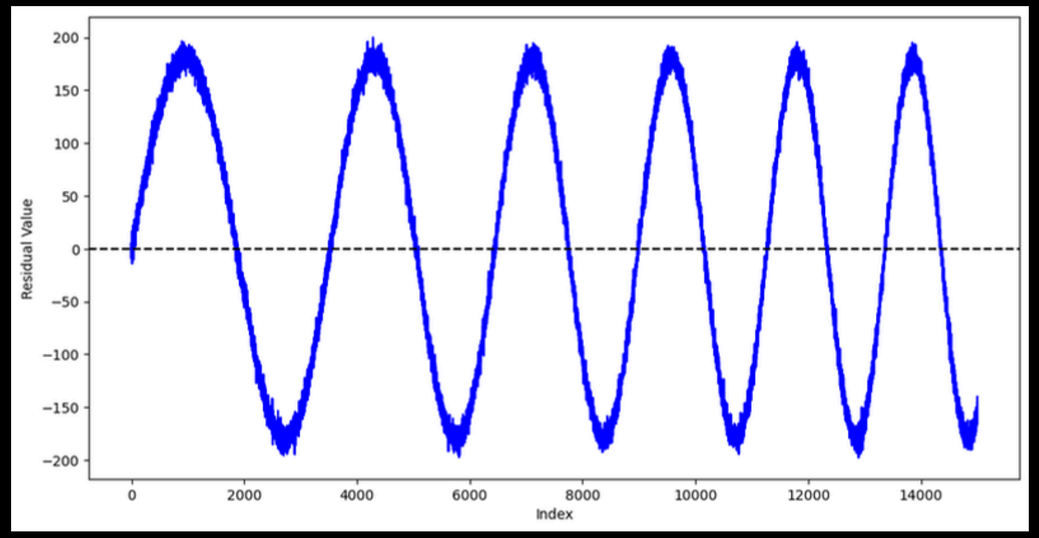


No Trade vs Double Differentiation of log of Asset1 ask_px1

EXPLORATORY DATA ANALYSIS

02 ARBITRAGE ANALYSIS

- We also found a **high correlation** between **asset2, asset4 and asset7**
- The correlation between the sum of the order book prices of asset2 & asset4 & order book prices of asset7 was greater than 0.9
- The difference of order book data of asset7 & order book data of asset2 & asset4 generated a sin curve with increasing frequency.
- We built a strategy around mean reverting pattern to generate signals for asset2, asset4 and asset7.



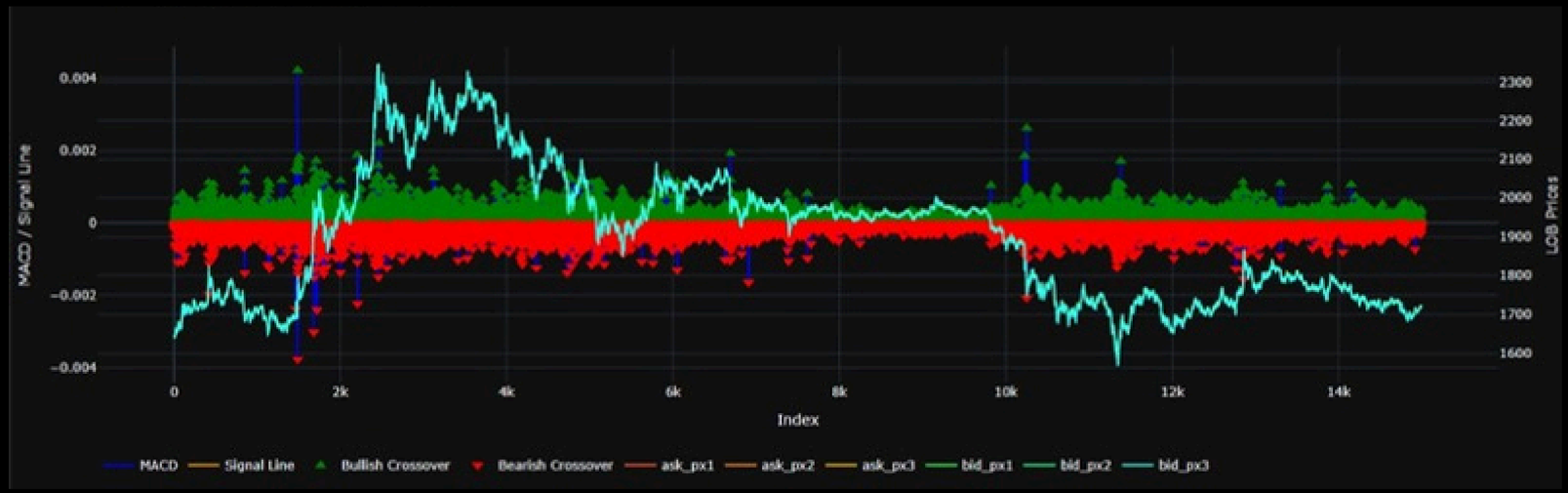
Plot of bid_px1 of Asset7 - (Asset2 + Asset4)

TRADING STRATEGIES

STRATEGY FOR ASSEST 1

- During the initial Exploratory Data Analysis, we found that the Notrade signal provided in the PS was highly correlated with the prices of Asset 1.

- Using this information, we used MACD on the Signal column in the Notrade dataset, to detect price momentum to identify trading points.



STRATEGY FOR ASSESTS 3 & 6

STOCHASTIC RESIDUAL SPREAD

Pairs trading is a market-neutral strategy, exploiting temporary mispricings between two related financial instruments, based on the premise that their relative valuation, or ”spread,” is mean-reverting. We are using a custom made model inspired from a research paper.

The **Stochastic Residual Spread** model, a sophisticated strategy that elevates pairs trading from simple heuristics to a robust econometric framework grounded in modern asset pricing theory. Its core in innovations are:

- Defining the spread as a residual in returns after accounting for systematic risk.
- Using a State-Space Model and the Kalman Filter to estimate the unobservable (latent) true mispricing, systematically separating signal from noise.

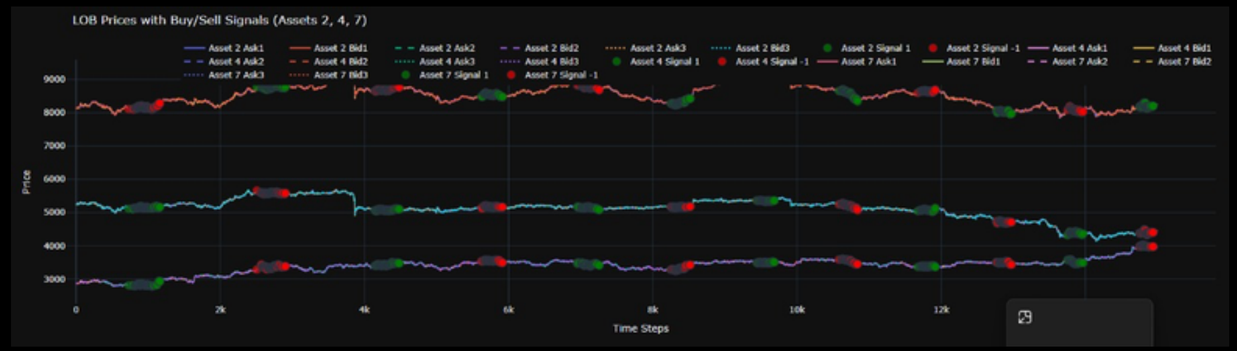
STRATEGY FOR ASSESTS 2,4 & 7

STATISTICAL ARBITRAGE

- This is a market- neutral trading strategy exploiting the mispricings between the related assets, here being assets 2, 4 and 7.
- From the EDA of Assets 2, 4 and 7, Residual plots show clear oscillatory, mean- reverting patterns around zero. This validates that tri-asset residuals are predictable and can be systematically exploited
- The strategy that we went for was:

$$R_t = P_t^{(7)} - P_t^{(4)} - P_t^{(2)}$$

  - If $R_t >$ Threshold: Short 7, Long 2 and 4.
  - If $R_t <$ Threshold: Long 7, Short 2 and 4.



STRATEGY FOR ASSEST 5

CANDLESTICK

We created custom candlesticks by leveraging data from the order book and trade logs. Here's how we derived the candlestick elements:

- High (VWAP of Ask Prices):
  The VWAP of the ask prices was used to represent the high of the candlestick.

- Low (VWAP of Bid Prices):
  The VWAP of the bid prices was used to represent the low of the candlestick.

- Open (Last Executed Trade Price):
  The open of the candlestick was set to the VWAP from the trade log of the prior datapoint.

- Close (Based on Order Book Imbalance - OBI):
  The close was determined based on the Order Book Imbalance (OBI):
  If OBI > threshold, the close is the best bid.
  If OBI < -threshold, the close is the best ask.
  Otherwise, the close is the mid-price.

CANDLESTICK

CANDLE STICK FOR ASSET5

MOMENTUM - EQUILIBRIUM STRATEGY

This trading strategy relies upon the signal generated by an **Momentum - Equilibrium framework**. We use the generated candlesticks to generate 4 parameters, to generate signals:

- Momentum Conductor: (max(High9) + min(Low9))/2
- Equilibrium Baseline: (max(High26) + min(Low26))/2
- Alpha & Beta Projection Lines: Two forward-projected lines that form the dynamic Equilibrium Zone.

For the signal generation part:

- A Buy Signal (+1) is generated when the Momentum Conductor crosses above the Equilibrium Baseline.
- A Sell Signal (-1) is generated when the Momentum Conductor crosses below the Equilibrium Baseline.
- A Neutral Signal (0) is registered when there is no crossover

MOMENTUM - EQUILIBRIUM STRATEGY

We are using a **Dynamic Volume Sizing,** by measuring a continuous score from 0 to 1 is calculated to measure the confidence behind the signal. It's based on a metric known as Equilibrium Strength :

It measures the trend's force by the price's normalized distance from the Equilibrium Zone : StrengthFinal = (StrengthEq)

The final trade volume is a direct function of the conviction score, scaled by a maximum position limit, ensuring that higher-conviction signals receive more capital.

Volume = floor(StrengthFinal × max_pos)

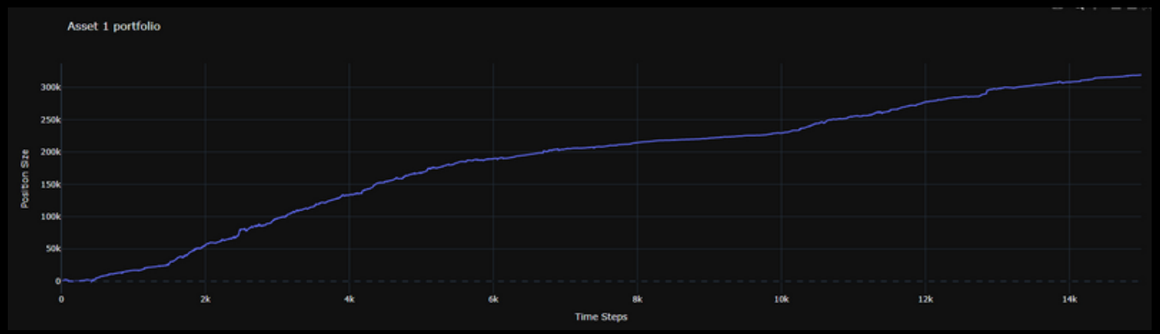
BACKTESTING

Class Backtesting :

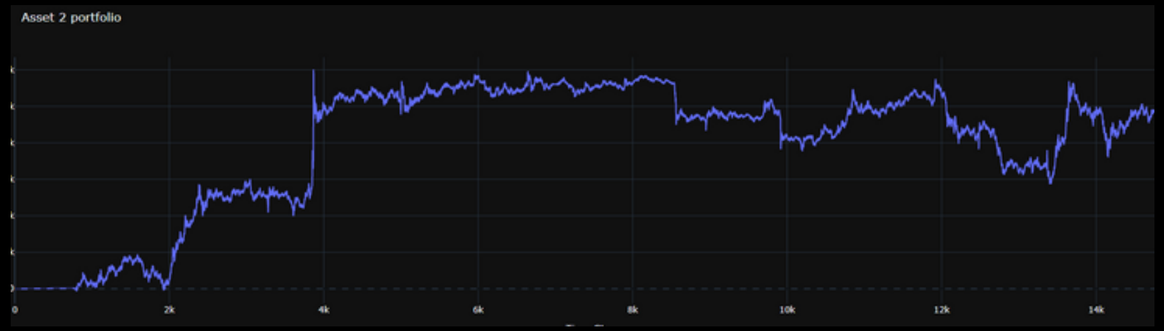
- def execute_trade():
  Executes the signal starting from the best possible price and moving to the next if volume requirement for strategy exceeds available liquidity for that price on that index.
- def run_backtest():
  Calls the execute_trade function, sends the signal along with volume for each asset .
- def plot_positions():
  Plots the positions we are taking on the different assets at each timestamp
- def plot_portfolio():
  Plots the portfolio value at each timestamp.
- def check_stop_loss():
  Implements stop loss on the portfolio to minimize drawdown.

RESULTS

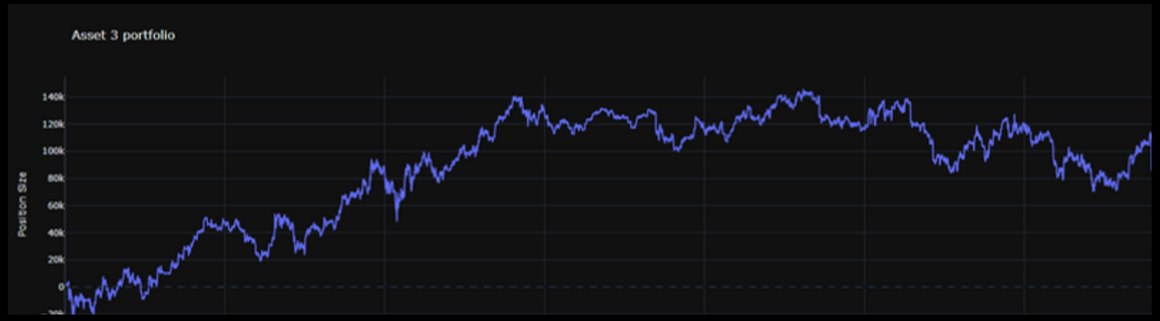
TOTAL PNL = 14,46,515.50



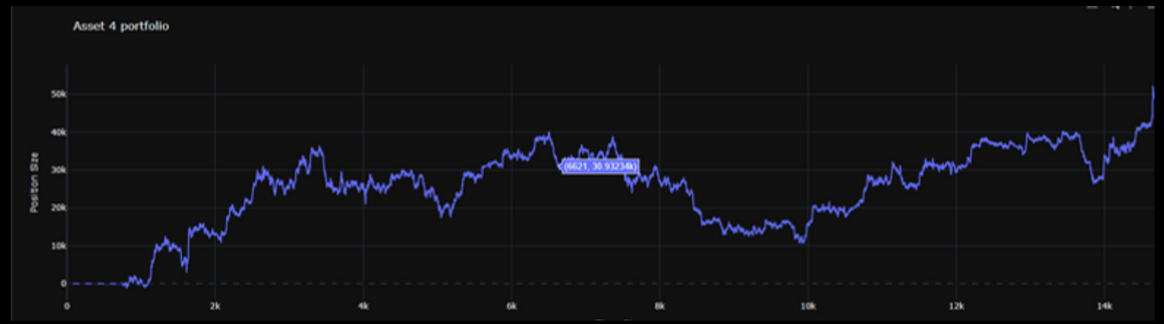
PNL_ASSET1: 3,19,067.78



PNL_ASSET2: 44,146.05

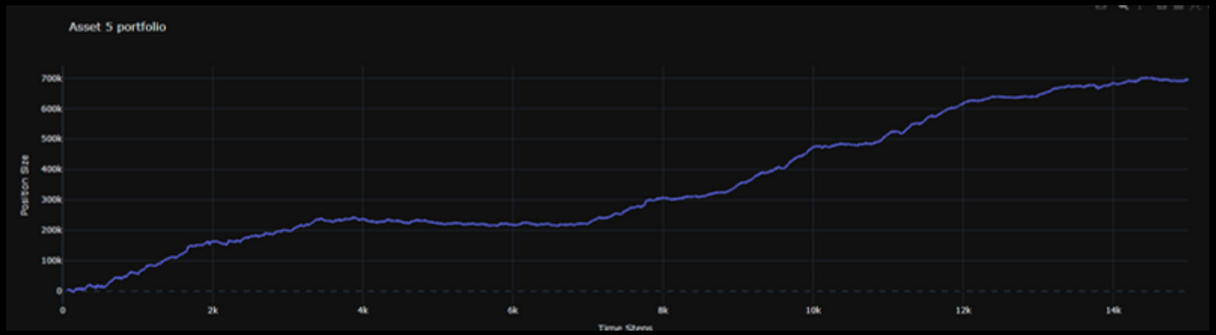


PNL_ASSET3: 1,43,347.28

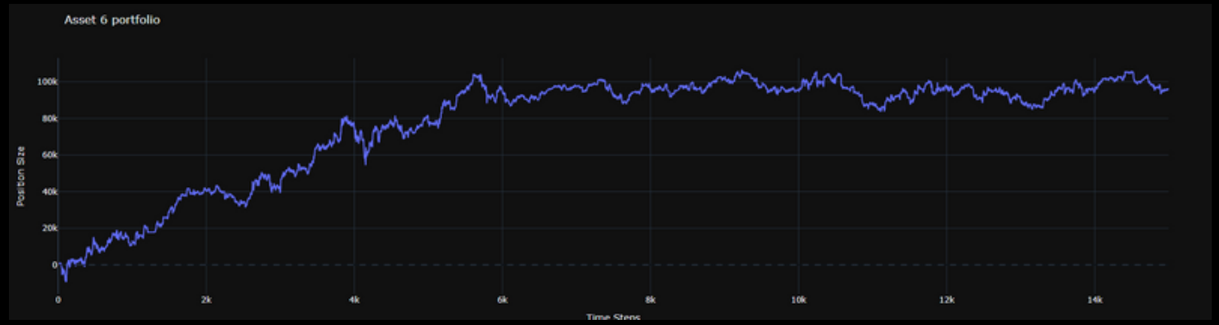


PNL_ASSET4: 44,976.53

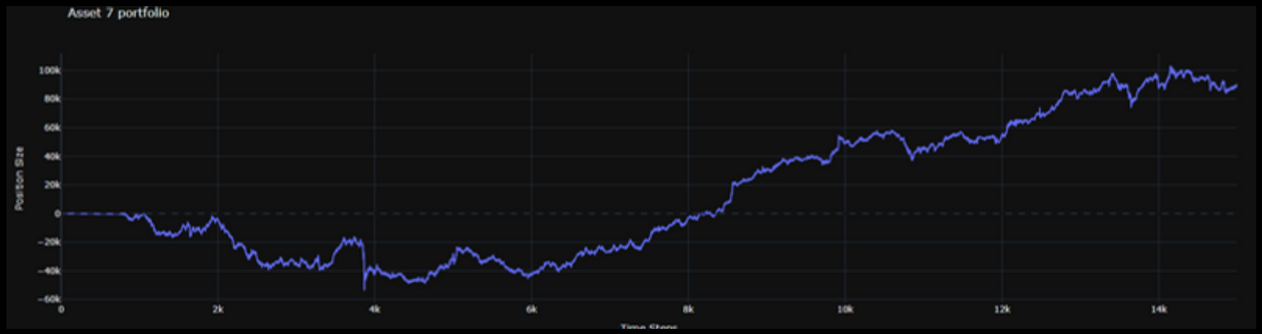
BOOK ORIENTED LIMIT TRADING



PNL_ASSET5: 6,95,864.48



PNL_ASSET6: 1,05,363.90



PNL_ASSET7: 93,749.47

THANK YOU